Programming Challenges

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1. `showChar` Method

Write a method named `showChar`. The method should accept two arguments: a reference to a `String` object and an integer. The integer argument is a character position within the `String`, with the first character being at position 0. When the method executes, it should display the character at that character position. Here is an example of a call to the method:

```
showChar("New York", 2);
```

In this call, the method will display the character `w` because it is in position 2. Demonstrate the method in a complete program.

2. Retail Price Calculator

VideoNote The Retail Price Calculator Problem

Write a program that asks the user to enter an item's wholesale cost and its markup percentage. It should then display the item's retail price. For example:

- If an item's wholesale cost is 5.00 and its markup percentage is 100 percent, then the item's retail price is 10.00.
- If an item's wholesale cost is 5.00 and its markup percentage is 50 percent, then the item's retail price is 7.50.

The program should have a method named `calculateRetail` that receives the wholesale cost and the markup percentage as arguments, and returns the retail price of the item.

3. Rectangle Area—Complete the Program

If you have downloaded the book's source code from www.pearsonhighered.com/gaddis, you will find a partially written program named `AreaRectangle.java` in this chapter's source code folder. Your job is to complete the program. When it is complete, the program will ask the user to enter the width and length of a rectangle, and then display the rectangle's area. The program calls the following methods, which have not been written:

- `getLength`—This method should ask the user to enter the rectangle's length, and then return that value as a `double`.
- `getWidth`—This method should ask the user to enter the rectangle's width, and then return that value as a `double`.
- `getArea`—This method should accept the rectangle's length and width as arguments, and return the rectangle's area. The area is calculated by multiplying the length by the width.
- `displayData`—This method should accept the rectangle's length, width, and area as arguments, and display them in an appropriate message on the screen.

4. Paint Job Estimator

A painting company has determined that for every 115 square feet of wall space, one gallon of paint and eight hours of labor will be required. The company charges $18.00 per hour for labor. Write a program that allows the user to enter the number of rooms to be painted and the price of the paint per gallon. It should also ask for the square feet of wall space in each room. The program should have methods that return the following data:

- The number of gallons of paint required
- The hours of labor required

- The cost of the paint
- The labor charges
- The total cost of the paint job

Then it should display the data on the screen.

5. Falling Distance

When an object is falling because of gravity, the following formula can be used to determine the distance the object falls in a specific time period:

$$d = \frac{1}{2} g t^2$$

The variables in the formula are as follows: d is the distance in meters, g is 9.8, and t is the amount of time, in seconds, that the object has been falling.

Write a method named `fallingDistance` that accepts an object's falling time (in seconds) as an argument. The method should return the distance, in meters, that the object has fallen during that time interval. Demonstrate the method by calling it in a loop that passes the values 1 through 10 as arguments, and displays the return value.

6. Celsius Temperature Table

The formula for converting a temperature from Fahrenheit to Celsius is

$$C = \frac{5}{9}(F - 32)$$

where F is the Fahrenheit temperature and C is the Celsius temperature. Write a method named `celsius` that accepts a Fahrenheit temperature as an argument. The method should return the temperature, converted to Celsius. Demonstrate the method by calling it in a loop that displays a table of the Fahrenheit temperatures 0 through 20 and their Celsius equivalents.

7. Test Average and Grade

Write a program that asks the user to enter five test scores. The program should display a letter grade for each score and the average test score. Write the following methods in the program:

- `calcAverage`—This method should accept five test scores as arguments and return the average of the scores.
- `determineGrade`—This method should accept a test score as an argument and return a letter grade for the score, based on the following grading scale:

Score	Letter Grade
90–100	A
80–89	B
70–79	C
60–69	D
Below 60	F

8. Conversion Program

Write a program that asks the user to enter a distance in meters. The program will then present the following menu of selections:

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

The program will convert the distance to kilometers, inches, or feet, depending on the user's selection. Here are the specific requirements:

- Write a `void` method named `showKilometers`, which accepts the number of meters as an argument. The method should display the argument converted to kilometers. Convert the meters to kilometers using the following formula:

  ```
  kilometers = meters * 0.001
  ```
- Write a `void` method named `showInches`, which accepts the number of meters as an argument. The method should display the argument converted to inches. Convert the meters to inches using the following formula:

  ```
  inches = meters * 39.37
  ```
- Write a `void` method named `showFeet`, which accepts the number of meters as an argument. The method should display the argument converted to feet. Convert the meters to feet using the following formula:

  ```
  feet = meters * 3.281
  ```
- Write a `void` method named `menu` that displays the menu of selections. This method should not accept any arguments.
- The program should continue to display the menu until the user enters 4 to quit the program.
- The program should not accept negative numbers for the distance in meters.
- If the user selects an invalid choice from the menu, the program should display an error message.

Here is an example session with the program, using console input. The user's input is shown in bold.

```
Enter a distance in meters: 500 [Enter]
1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

Enter your choice: 1 [Enter]
500 meters is 0.5 kilometers.

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program
```

```
Enter your choice: 3 [Enter]
500 meters is 1640.5 feet.

1. Convert to kilometers
2. Convert to inches
3. Convert to feet
4. Quit the program

Enter your choice: 4 [Enter]
Bye!
```

9. Distance Traveled Modification

The distance a vehicle travels can be calculated as follows:

*Distance = Speed * Time*

Write a method named `distance` that accepts a vehicle's speed and time as arguments, and returns the distance the vehicle has traveled. Modify the "Distance Traveled" program you wrote in Chapter 4 (Programming Challenge 2) to use the method.

10. Stock Profit

The profit from the sale of a stock can be calculated as follows:

$$Profit = ((NS \times SP) - SC) - ((NS \times PP) + PC)$$

where *NS* is the number of shares, *PP* is the purchase price per share, *PC* is the purchase commission paid, *SP* is the sale price per share, and *SC* is the sale commission paid. If the calculation yields a positive value, then the sale of the stock resulted in a profit. If the calculation yields a negative number, then the sale resulted in a loss.

Write a method that accepts as arguments the number of shares, the purchase price per share, the purchase commission paid, the sale price per share, and the sale commission paid. The method should return the profit (or loss) from the sale of stock. Demonstrate the method in a program that asks the user to enter the necessary data and displays the amount of the profit or loss.

11. Multiple Stock Sales

Use the method that you wrote for Programming Challenge 10 (Stock Profit) in a program that calculates the total profit or loss from the sale of multiple stocks. The program should ask the user for the number of stock sales, and the necessary data for each stock sale. It should accumulate the profit or loss for each stock sale and then display the total.

12. Kinetic Energy

In physics, an object that is in motion is said to have kinetic energy. The following formula can be used to determine a moving object's kinetic energy:

$$KE = \frac{1}{2} mv^2$$

The variables in the formula are as follows: *KE* is the kinetic energy, *m* is the object's mass in kilograms, and *v* is the object's velocity, in meters per second.

Write a method named `kineticEnergy` that accepts an object's mass (in kilograms) and velocity (in meters per second) as arguments. The method should return the amount of kinetic energy that the object has. Demonstrate the method by calling it in a program that asks the user to enter values for mass and velocity.

13. `isPrime` Method

A prime number is a number that is evenly divisible only by itself and 1. For example, the number 5 is prime because it can be evenly divided only by 1 and 5. The number 6, however, is not prime because it can be divided evenly by 1, 2, 3, and 6.

Write a method named `isPrime`, which takes an integer as an argument and returns `true` if the argument is a prime number, or `false` otherwise. Demonstrate the method in a complete program.

> **TIP:** Recall that the `%` operator divides one number by another, and returns the remainder of the division. In an expression such as `num1 % num2`, the `%` operator will return 0 if `num1` is evenly divisible by `num2`.

14. Prime Number List

Use the `isPrime` method that you wrote in Programming Challenge 13 in a program that stores a list of all the prime numbers from 1 through 100 in a file.

15. Even/Odd Counter

You can use the following logic to determine whether a number is even or odd:

```
if ((number % 2) == 0)
{
   // The number is even.
}
else
{
   // The number is odd.
}
```

Write a program with a method named `isEven` that accepts an `int` argument. The method should return `true` if the argument is even, or `false` otherwise. The program's `main` method should use a loop to generate 100 random integers. It should use the `isEven` method to determine whether each random number is even, or odd. When the loop is finished, the program should display the number of even numbers that were generated, and the number of odd numbers.

16. Present Value

Suppose you want to deposit a certain amount of money into a savings account, and then leave it alone to draw interest for the next 10 years. At the end of 10 years, you would like to have $10,000 in the account. How much do you need to deposit today to make that happen? You can use the following formula, which is known as the present value formula, to find out:

$$P = \frac{F}{(1 + r)^n}$$

The terms in the formula are as follows:

- P is the **present value**, or the amount that you need to deposit today.
- F is the **future value** that you want in the account. (In this case, F is $10,000.)
- r is the **annual interest rate**.
- n is the **number of years** that you plan to let the money sit in the account.

Write a method named `presentValue` that performs this calculation. The method should accept the future value, annual interest rate, and number of years as arguments. It should return the present value, which is the amount that you need to deposit today. Demonstrate the method in a program that lets the user experiment with different values for the formula's terms.

17. Rock, Paper, Scissors Game

Write a program that lets the user play the game of Rock, Paper, Scissors against the computer. The program should work as follows.

1. When the program begins, a random number in the range of 1 through 3 is generated. If the number is 1, then the computer has chosen rock. If the number is 2, then the computer has chosen paper. If the number is 3, then the computer has chosen scissors. (Don't display the computer's choice yet.)
2. The user enters his or her choice of "rock", "paper", or "scissors" at the keyboard. (You can use a menu if you prefer.)
3. The computer's choice is displayed.
4. A winner is selected according to the following rules:

- If one player chooses rock and the other player chooses scissors, then rock wins. (The rock smashes the scissors.)
- If one player chooses scissors and the other player chooses paper, then scissors wins. (Scissors cuts paper.)
- If one player chooses paper and the other player chooses rock, then paper wins. (Paper wraps rock.)
- If both players make the same choice, the game must be played again to determine the winner.

Be sure to divide the program into methods that perform each major task.

18. ESP Game

Write a program that tests your ESP (extrasensory perception). The program should randomly select the name of a color from the following list of words:

Red, Green, Blue, Orange, Yellow

To select a word, the program can generate a random number. For example, if the number is 0, the selected word is *Red*; if the number is 1, the selected word is *Green*; and so forth.

Next, the program should ask the user to enter the color that the computer has selected. After the user has entered his or her guess, the program should display the name of the randomly selected color. The program should repeat this 10 times and then display the number of times the user correctly guessed the selected color. Be sure to modularize the program into methods that perform each major task.